Spatial Trust Primitives

Composable Location Proofs for the Embodied AI Era

ARIA Scaling Trust • Track 3.2: Cyber-Physical Primitives (Exploratory Project) • 6 months • ~£125-150k
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Section 0: Summary

There is no formal framework for verifiable location claims between agents. Location data is ubiquitous, and almost entirely unverified. GPS coordinates can be spoofed with commodity hardware. There is no standard for composing evidence from heterogeneous sources, no theory for modeling independence between signals, and no method for quantifying forgery cost. As embodied AI systems enter governed physical spaces at scale, this gap becomes a critical vulnerability: "unforgeable receipts for robot actions" and "system-level consensus for cyber-physical systems" both require location evidence that agents can evaluate and trust.

We propose to formalize **location proofs** as a cyber-physical primitive (Track 3.2). A location proof bundles a spatial claim with evidence collected from independent proof-of-location systems - e.g. GNSS, network latency, NFC proximity, environmental fingerprinting, institutional attestation - in a standardized, machine-readable format. The primitive is compositional: stamps from uncorrelated systems combine into a proof whose forgery requires simultaneous, consistent deception across independent channels. It is also unopinionated: the framework standardizes evidence structure without prescribing how verifiers weigh it, because forgery cost is contextual to the interaction and the risk tolerance of the parties involved.

We have a working research implementation. The Astral Protocol includes an SDK with location proof plugins producing signed stamps from multiple real PoL systems, and a TEE-hosted verification service producing signed attestations without exposing raw coordinates. What is missing - and what this project delivers - is the foundation for a field that does not yet have a formal vocabulary. We would like to produce:

- A **taxonomy of proof-of-location systems**: rigorous classification across system families, with independence characterization and forgery cost profiles. This identifies the irreducible dimensions that inform credibility assessment.
- A **composition theory**: formal treatment of how heterogeneous location evidence combines, when sources provide independent versus redundant information, and what mathematical frameworks (Dempster-Shafer, Bayesian, information-theoretic) are appropriate.
- A **durability affordances framework**: systematic mapping of the levers - intrinsic (hardware attestation, signal authentication, secure peripherals) and extrinsic (legal liability, economic stakes, reputational cost) - available to raise forgery cost, and how they interact.

The PI has worked on verifiable geospatial systems for seven years. The timing - embodied AI deploying now, formal standards not yet set - makes this the moment to establish the foundations.

Section 1: Programme and Technical

Motivation

"Where are you?" It's one of the first questions asked in any coordination task involving the physical world. And yet, there is no reliable way to make credible claims about the location of devices, people, assets or events.

Navigating the world safely, and securely interacting with others, requires a degree of spatial awareness at the local, regional and global levels. Physical sensors supply signals required for positional self-awareness: with LIDAR, optical cameras, GNSS receivers, accelerometers, and other sensors, machines can sense with sufficient accuracy and precision to transit dense and messy urban and natural environments like city streets (Waymo, 2024) and forests (Loquercio et al., 2021). Intra-agent positional awareness capabilities are well-developed and rapidly improving.

Inter-agent location reporting - especially when the verifying agent has reason to be skeptical about the veracity of the location claim - has been explored in the past decades (Saroiu & Wolman, 2009; Chandran et al., 2009; Foamspace Corp., 2018; Zafar & Khan, 2016), but verifiable positioning systems and reporting protocols are still in their infancy. Most prior work has focused on building individual proof-of-location systems around a single signal type. None has engaged with the practical core of the problem: location claims are a decision tool, contextual to the nature of the interaction and the risk tolerance of the verifier.

As embodied AI agent populations enter physical spaces at scale, urgency to develop the capability to coordinate around believable location claims is growing. Location is perhaps the most pervasive and least formalized component of cyber-physical trust. In one sense, a huge portion of what it means to verify a cyber-physical process *is* location verification: did the robot perform this procedure? That's largely a question of whether the right components were in the right places at the right times. Did the delivery happen? Did the sensor reading come from the protected area? Did the drone enter restricted airspace? The capability to make credible claims about location to skeptical counterparties is an essential element in building "unforgeable receipts for robot actions" and "system-level consensus for cyber-physical systems", as framed by the ARIA Scaling Trust thesis (Obadia and Greco, 2026).

Yet technically, little effort has been put towards establishing secure protocols for inter-agent communication about where things happen, and even less towards verification in adversarial contexts. When there are incentives to lie - to access privileged information, get paid, avoid liability, enter restricted areas - how is the position and timing of an event verified? The potential of such “position-based cryptography” and verifiable location-based services is enormous in areas as divergent as cybersecurity, environmental management, transport systems, fraud and compliance, risk, and geopolitical strategy. But the infrastructure required to build such next-generation location-aware verifiable applications depends heavily on robust location verification technology.

We seek ARIA funding to rigorously formalize fundamental research into an underdeveloped cyber-physical primitive: location proofs (Track 3.2), a key enabling technology for robust location-aware coordination of multi-agent cyber-physical systems.

“Proving” Location

Location reporting in online systems is ubiquitous. GeoIP, mobile OS location services, address verification, NFC scans - countless applications modify their logic based on the inferred or reported location of their users. But how well do these claims hold up?

The current answer is surprisingly weak. In most cases, location data can be edited freely or forged easily. IP addresses can be manipulated with consumer VPNs; GPS coordinates can be rewritten in developer tools; GNSS signals can be spoofed using commodity software-defined receivers worth a few hundred dollars (Humphreys et al., 2008). In adversarial contexts where there is an incentive to lie about location, there are no widely-adopted protocols or patterns that make location claims legible and credible to a mistrusting, uncoordinated verifier.

Table 1: Verifiable Location Reporting Present and Future

Question	Current state	Goal
How does an agent describe where something happened?	A wide range of formats, coordinate systems, projections, and conventions. No interoperability - agents built by different teams can't reliably interpret each other's spatial claims.	A self-describing, machine-readable format for location claims that any agent can interpret regardless of who built it.
How does an agent know whether to believe a location claim?	Trust is implicit or binary. No rigorous way to assess how hard it would be to forge.	A multidimensional assessment of evidence quality - how accurate, how independent, how resistant to forgery - that a security reasoner can evaluate against its own policy.
What happens when multiple sources of evidence are available?	Each system produces its own output in its own format. No standard way to combine or compare them. Sensor fusion is proprietary and opaque.	A formal framework for composing heterogeneous evidence, modeling which sources are genuinely independent, and producing a unified assessment.
How hard is it to lie about where something happened?	Largely unquantified. Consumer VPNs are ubiquitous, GPS spoofing equipment costs ~$300. No systematic way to reason about forgery difficulty or relate it to the value at stake.	Forgery cost characterized across technical, economic, and social dimensions - legible enough that an agent can decide whether the evidence is strong enough for the interaction.
How can a claim be verified without trusting the claimant?	Very limited options: IP address lookup, perhaps mobile device signatures. Little cryptographic verification of location evidence.	Verifiable artifacts with digital signatures, reproducible evaluation, and privacy-preserving verification - the claimant proves where they were without revealing exact coordinates.

Location proofs as a compositional primitive

To address this gap, we have been developing a framework for the creation and verification of multi-factor location proofs. For clarity: we are not designing new proof-of-location (PoL) systems; instead, we are focused on how evidence collected from different PoL systems can be composed and organized into machine-readable, user-controlled verifiable artifacts.

Our framework is oriented around *raising the cost of lying* about where things happened, a significant reframing when compared to prior work. Location claims are important parameters in such a wide range of decisions, it is inappropriate to put opinionated emphasis on a specific location verification technique. Instead, the primitive should accommodate as wide a set of use cases as possible, and empower system developers and verifiers to make informed, legible decisions about the forms of evidence they're willing to accept in their context. As a principle, the cost of forging a location proof should exceed the value of the transaction it underpins.

A ***location proof*** is a digital artifact that describes the location and timing of an identifiable event (the ***location claim***), bundled with evidence supporting the claim that would require technical manipulation, collusion, or fraud to forge (the ***location evidence***). The claim is a tuple *(x, L, T)* - entity *x* was in region *L* during time interval *T*.

The evidence is created from observable ***signals*** collected from independent proof-of-location systems processed into verifiable ***location stamps***. Our initial survey has identified seven families of PoL systems, classified by the nature of the enabling physical or social phenomenon.

Differentiation

By stepping a level above individual PoL systems, and focusing on composing and verifying location evidence from an open-ended range of PoL systems, we address multiple unaddressed problems at once. The first is the compatibility problem: parsing and processing evidence from different systems requires custom integrations; a protocol establishes clear data formats and processes for production and consumption. Much more interestingly, we are also paving the way for harder *multi-factor location proofs*, which compose location stamps from uncorrelated PoL systems. Forgery costs scale non-linearly as a function of the number and diversity of PoL systems a malicious agent would need to simultaneously manipulate.

For instance, to forge an example multi-factor location proof would require that an attacker simultaneously spoof GPS signals, fake WiFi fingerprints, forge an NFC proximity check, and manipulate network latency measurements - each requiring different equipment, different expertise, and physical co-presence that conflicts with remote attack vectors. The difficulty of maintaining a coherent lie across independent channels grows faster than the sum of individual forgery costs.

Our vision is for a library of location proof plugins - components available to agents that allow them to connect to relevant PoL systems to collect signals, process them into stamps, and output location proofs. Of course, the compatible plugins depend on sensors, hardware, and software capabilities of the agent, which will in turn impact the “hardness” of the location proof an agent can produce.

Location proof evaluation

So, a location proof is a location claim plus pieces of supporting evidence collected from one or more proof-of-location systems. A verifier can then evaluate how well the evidence supports the claim, and assess how difficult it would be to forge the full evidence set.

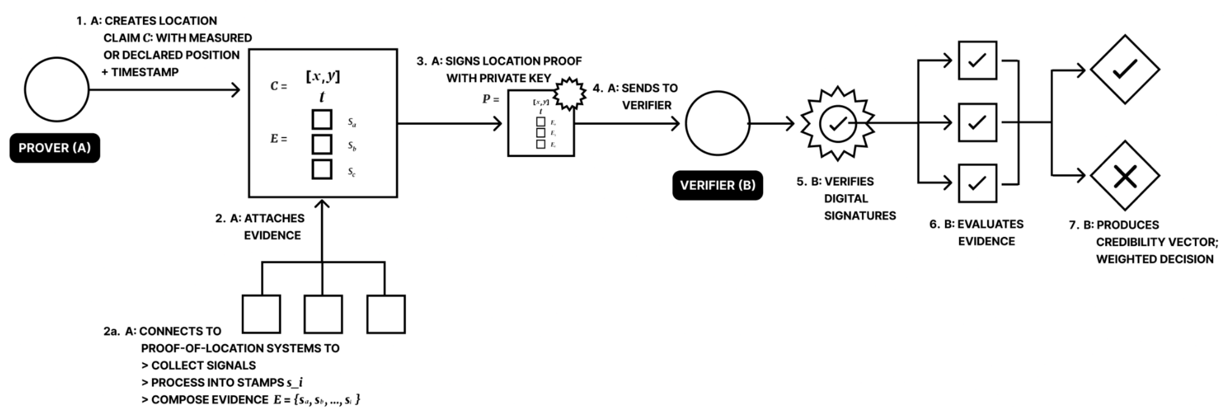
More precisely, a verifier processes the location proof {C, E} with an ***evaluation function***, which considers multiple factors when evaluating evidence against a claim:

- **Robustness:** the intrinsic reliability of each contributing system, including calibration accuracy, error bounds, and resistance to forgery or manipulation.
- **Independence:** the degree to which constituent systems provide distinct information rather than redundant signals.
- **Fit to the claim:** how directly the collected evidence pertains to the specific entity, region, and time interval asserted.

Importantly, the evaluation function is a variable in the framework. Verifiers will vary in the value they place on different evidence sources - some may, for example, place great weight on evidence supplied by “official” sources like government bodies, while in other contexts such evidence may do nothing to increase trust. Again, the framework is designed to be unopinionated: we don’t take a position on what is true, we just make the evidence legible so system designers can make more informed decisions.

A more thorough analytical sketch of our framework can be found in our October 2025 post *Towards Harder Location Proofs* (Hoopes, 2025).

Figure 1: Location Proof Lifecycle



Verifiable location-based services

On their own, location proofs improve trust by introducing rigor to location reporting. However, proving location only partially answers the actual questions being asked, which often involve discerning spatiotemporal relationships. Did this ship enter a Marine Protected Area? Is this request coming from an authorized country? How long was this drone in the congestion zone?

Verifiable location reports are a necessary element to answer these questions, but insufficient. We also need to access some kind of reference geographic feature - most often a polygon, a point and radial buffer, or a bounding box - and the ability to verifiably compute the topological relationship between the prover and the reference feature. Verifiable location-based services open a wide new range of applications to support cyber-physical coordination.

What we have built

We have built v0 of the Astral Protocol, a platform for verifiable location-based services. The system includes: location proof plugins producing signed location stamps from multiple PoL systems; an orchestration SDK composing stamps into signed location proofs; a TEE-hosted verification endpoint that evaluates location evidence without exposing raw coordinates; and a TEE-hosted geospatial compute endpoint for spatial policy evaluation. All signed artifacts support both offchain and onchain distribution.

The codebase is open and operational: approximately 5,000 lines across the SDK and services, processing real stamps from real proof-of-location systems through a real verification pipeline.

We are active in the Open Geospatial Consortium (Blockchain and DLT Working Group; observer, Integrity, Provenance and Trust Working Group). Pilot users are building on the platform, including a team developing Topocurrencies (Rawson and Borreani, 2026) and a FFDW-funded collaboration with the University of Maryland integrating verifiable location data into a flood mapping application. Our PI is a Research Affiliate at UMD and is supporting delivery of the FFDW grant.

Technical risks and unknowns

Correlation modeling. Measuring independence between PoL systems that may share partial signal overlap - GPS-derived stamps with different processing chains, or WiFi positioning calibrated against GPS ground truth - requires a formal framework that does not yet exist. Fallback: empirical correlation measurement using controlled experiments with known ground truth.

Credibility vector dimensionality. The irreducible dimensions may be context-dependent rather than universal. If principled dimensionality reduction proves elusive, the fallback is domain-specific credibility profiles rather than a single canonical vector.

Normalization. A 3-meter GPS fix, a city-level network latency proof, and a binary NFC proximity check operate at fundamentally different spatial scales with different semantics. Making these commensurable without collapsing meaningful distinctions is a genuine calibration challenge.

Signal supply chain. The strength of a location proof is bounded by the weakest link between physical signal and signed output. Pushing verifiability toward the signal source (secure peripherals, authenticated signals like Galileo OSNMA) raises forgery cost substantially, but depends on hardware decisions by chipset manufacturers and OEMs largely outside our influence. The framework must work with today's hardware while accommodating stronger guarantees as they emerge.

Ecosystem dependence. The highest-value primitives - authenticated satellite signals, secure sensor attestation - require upstream decisions by space agencies, chipset vendors, and device manufacturers. We define interfaces and demonstrate value; we cannot compel adoption. ARIA's programme structure mitigates this: cross-track collaboration with teams building new physical trust anchors provides integration paths that don't depend on external ecosystem adoption.

Research agenda: formalizing the primitive

The focus of this proposed work is to establish the formal foundations for location proofs as a rigorous cyber-physical primitive.

Taxonomy of proof-of-location systems. We have conducted an inclusive survey of proof-of-location systems, organizing them into seven categories based on their underlying trust mechanisms and forgery models:

1. **Authority-based.** Mobile network operators, government registries, institutional attestations. Trust derives from the authority's reputation.
2. **Social / peer witness.** Co-location attestation by other agents or humans. Trust derives from the independence and honesty of witnesses.
3. **Near-field machine.** NFC, Bluetooth Low Energy, Ultra-Wideband ranging, distance-bounding protocols. Trust derives from physics - radio propagation constraints. Many PUFs may sit here.
4. **Network multilateration.** WiFi RTT, cellular TDOA, network latency proofs (e.g. WitnessChain). Trust derives from the difficulty of simultaneously spoofing multiple independent network paths.
5. **Sensor data.** GNSS (including authenticated signals via Galileo OSNMA), barometric pressure, magnetic field fingerprints, audio environment signatures. Trust derives from the difficulty of fabricating consistent multi-sensor readings. Many PUFs could be classified here.
6. **Legal.** Notarized attestations, jurisdiction-derived claims, address registrations. Trust derives from legal accountability.
7. **Delegated.** Trusted third-party assertions - Find My network, assisted GPS, carrier-provided location. Trust derives from the service provider's infrastructure and incentives.

This taxonomy needs to be made rigorous. The first research output is a systematic survey that identifies the irreducible dimensions along which proof-of-location systems vary - the dimensions that will inform the design of the credibility vector output by evaluation functions. Each PoL system has a distinct forgery cost profile and associated durability affordances, and the critical insight is this: combining stamps from *independent* categories raises the difficulty of forgery because the attack vectors are physically unrelated. This work will formalize patterns across, and unique to, PoL systems, and provide a systematic approach to assessing and onboarding new location proof plugins.

Composition theory. How does location evidence compose? Two GPS-derived stamps share a root signal - if an attacker spoofs that signal, both are compromised simultaneously. But a GPS stamp and an NFC proximity check are independent - different physical phenomena, different attack surface. The composition theory must formally characterize:

- When two stamp types provide genuinely independent evidence versus redundant signals from a shared or correlated root
- How independent evidence combines to raise the difficulty of forgery
- What mathematical formalisms are appropriate: Dempster-Shafer belief functions, Bayesian networks, information-theoretic approaches, or something new

This is open research. Rigorously quantifying the independence between heterogeneous proof-of-location systems is, as far as we can determine, an unsolved problem. We plan to investigate information-theoretic approaches (mutual information between stamp outputs given shared inputs) and structural approaches (dependency graphs over signal sources).

Durability affordances. What are the levers available to raise the cost or difficulty of lying? These split into two categories:

Intrinsic affordances are properties of the signal and hardware - they apply universally regardless of context. Pushing cryptographic verifiability further up the information supply chain is the primary technical lever: signed OSNMA signals are harder to forge than unsigned GPS; secure peripherals (sensors cryptographically bound to a secure element) are harder to circumvent than unattested sensors; TEE-processed evidence carries hardware-backed guarantees that software alone cannot provide.

Extrinsic affordances are consequences imposed by context - they depend on who is involved and what is at stake. A legal affidavit carries forgery cost proportional to the penalties for perjury in the relevant jurisdiction. A reputational system imposes social cost - a corporate entity discovered issuing fraudulent location claims faces stock price impact, regulatory scrutiny, loss of partnerships. Economic bonds or insurance requirements create financial stakes.

This distinction matters because the same evidence bundle has different effective forgery costs in different settings. The evaluation framework must surface both intrinsic and extrinsic properties without prescribing how they are weighted - different verifiers will have different evaluation functions based on their risk models and trust requirements. We standardize the evidence structure so that evaluation becomes a tractable problem; we do not prescribe the evaluation itself.

Credibility vector formalization. The output of the evaluation function is a credibility vector - a multidimensional quantification of the evidence profile. Each dimension captures a distinct aspect of how well the evidence supports the claim. We have some idea of what dimensions this credibility vector will contain - spatial accuracy, temporal integrity, source independence, forgery resistance, privacy guarantees, some decentralization metric (i.e. Nakamoto coefficient for PoL systems). Identifying the irreducible attributes that emerge from the taxonomy is both a research question and a design decision. The credibility vector is to location evidence what a feature embedding is to high-dimensional data: a projection onto the dimensions that carry meaningful information about trustworthiness.

The evidence function maps a claim and evidence bundle to a credibility vector: $\mathcal{E}(\mathbf{C}, \mathbf{E}) \rightarrow \mathbf{V}$. This is a *family* of functions, not a single formula - each instance is specific to a combination of stamp types, their correlation structure, and the claim type. An application-specific weighting function w: $\mathbf{V} \rightarrow p \in [0,1]$ maps the credibility vector to a scalar for decision thresholds. The vector preserves information; the weighting function is where policy meets evidence.

Work packages and milestones (6 months)

WP1 - Taxonomy and classification (months 1-3)

Systematic survey and formal classification of proof-of-location systems.

- Rigorous review of PoL systems across all categories, with threat models, forgery cost profiles, and independence characterization.
- Challenge and refine classification: are the seven proposed categories appropriate?
- Identify irreducible dimensions along which PoL systems vary - these will inform the structure of the credibility vector
- Characterize pairwise independence/correlation structure between categories;
- Map durability affordances: intrinsic (hardware, cryptographic, physical) and extrinsic (legal, economic, social/reputational) levers for raising forgery cost

Deliverable: Taxonomy paper submitted for peer review. Key contribution: a holistic classification of an inclusive set of proof-of-location systems with initial characterization of their composition properties.

WP2 - Composition theory and credibility formalization (months 2-6)

Formalize how location evidence composes and how to quantify the result.

- Define the evidence function family $E(C, E) \rightarrow V$ with explicit correlation modeling
- Develop independence measurement framework: given two stamp types, quantify their shared vs. distinct information
- Evaluate candidate formalisms (Dempster-Shafer, Bayesian networks, information-theoretic) against real stamp data
- Formalize forgery cost quantification: what does "cost" mean across technical, economic, and social dimensions? How do intrinsic and extrinsic affordances compose?
- Formalize the credibility vector: identify dimensions, define normalization, characterize the relationship between evidence properties and vector components

Deliverable: Composition theory working paper with formal definitions and empirical validation against real stamp data. Foundation for a set of benchmark evaluation functions.

WP3 - Durability affordances (months 4-6)

- Systematic mapping of intrinsic durability levers across PoL system categories: hardware attestation chains, signal authentication (OSNMA), secure peripheral architectures, cryptographic binding
- Systematic mapping of extrinsic durability levers: legal liability structures, economic bonds, reputational mechanisms, regulatory penalties
- For each lever, characterize the relationship between the affordance and its contribution to forgery cost - how does adding a secure peripheral change the cost profile vs. adding a legal penalty?
- Produce a durability affordance framework that plugin developers can use to document the hardening properties of their PoL system

Deliverable: Durability affordances paper characterizing the full landscape of forgery cost levers - intrinsic and extrinsic - across PoL system categories, with a framework for assessing how each lever contributes to the overall hardness of a location proof.

Milestones

Month 2 (go/no-go). Taxonomy framework with at least 6 PoL system types formally characterized, including pairwise independence assessment for at least 3 pairs. If the independence characterization does not produce meaningful distinctions between pairs - if all pairs look equivalently correlated - this signals the formal approach needs revision. Decision point: continue or pivot.

Month 4. Composition theory working draft with formal evidence function definition, demonstrated on real stamp data from at least 2 systems in the existing SDK. Forgery cost framework defined, with at least one worked example (e.g., GPS + WiFi RTT) showing how intrinsic and extrinsic affordances contribute.

Month 6. Taxonomy paper submitted. Composition theory paper drafted. Durability affordances paper drafted. Benchmark requirements documented for Phase 2.

Phase 2 scope (intended second solicitation)

This 6-month project establishes the formal foundations. The natural follow-on:

- **Arena integration.** Build a location proof benchmark as a component-level challenge. Design adversarial scenarios for the Arena: can you fool a composed credibility assessment even when individual stamps verify correctly?
- **Reference datasets and hardware deployments.** Procure OSNMA GPS receivers, deploy sensor arrays with ground-truth positions, build labeled datasets of location proofs across the durability spectrum. This requires hardware that cannot be scoped in Phase 1.
- **Programmable cryptography investigation.** Explore ZK circuit representations of location proof verification and evaluation algorithms - where the practical boundary sits between TEE-based and fully cryptographic verification.

Intended impact

- **Local alignment.** Location-aware safety specifications - parameterizing what constitutes "aligned" behavior based on where an agent operates - depend on verifiable location as a precondition.
- **Governance of autonomous systems.** Provably-compliant spatial policies for drones, autonomous vehicles, and robotic systems operating in regulated physical spaces.
- **Machine-enforceable spatially-constrained agreements.** Location-contingent contracts and treaties where compliance can be verified cryptographically rather than adjudicated after the fact.

Section 2: The Team

John Hoopes - Principal Investigator (0.8 FTE)
Geospatial systems researcher and engineer with 15 years professional experience. MSc Spatial Data Science & Visualisation with distinction, UCL (Centre for Advanced Spatial Analysis), 2018. Dissertation on decentralized physical-state verification using trusted IoT sensors, connected to smart contracts. Former Ordnance Survey innovation researcher (UK national mapping agency). Co-founded Toucan Protocol with Adam Spiers, building smart contract infrastructure for environmental asset management. Founded Astral Protocol / Sophia Systems - six years building verifiable geospatial systems, public-goods funded throughout with no dilutive capital. Published the composable location proofs framework (Flashbots Collective Forum, 2025) and a verifiable geospatial web vision paper (GISRUK 2025). Built the working implementation described in this proposal: astral-sdk, astral-location-services, and the TEE-hosted verification pipeline. Active in the Open Geospatial Consortium (Blockchain and DLT Working Group; observer, Integrity, Provenance and Trust Working Group).

Adam Spiers - Research Engineer (0.4 FTE)
Senior systems architect with 25+ years in distributed systems engineering. BA Mathematics and Computation, Oxford University. Co-founded Toucan Protocol with the PI, designing smart contract infrastructure for environmental asset management. Senior Architect at Panther Protocol, where he contributed to protocol design including cryptographic and zero-knowledge proof elements. Currently Protocol Architect for Hypercerts (Protocol Labs). Seven years at SUSE as Senior Architect for cloud and high availability infrastructure, including leading implementation of hardware-based memory encryption for VMs using AMD SEV - directly relevant experience with trusted execution environments. Won a ZKP hackathon implementing elliptic curve operations in ZoKrates. Extensive open-source track record (top 10,000 F/OSS contributors globally). Brings deep systems architecture, ZK and TEE experience, and a long-standing existing working relationship with the PI.

Quantitative researcher - to hire (0.4 FTE)
Profile: MSc or PhD in applied mathematics, statistics, information theory, or signal processing. Comfortable with formal modeling frameworks (Dempster-Shafer, Bayesian networks, copula-based dependence modeling). Some exposure to sensor fusion - in robotics, defense, aerospace, or geospatial. Able to take qualitative frameworks and express them in formalism that survives peer review. The composition theory and credibility vector formalization require this expertise.
Hiring plan: Recruit within first month of project via existing networks and ARIA community in the geospatial, sensor fusion, and applied math communities. The role is well-defined and the scope (6-month contract, specific deliverables) should attract strong candidates.

Advisors and collaborators

- **Professor Taylor Oshan** (University of Maryland) - quantitative geography, spatial analysis methodologies. Taylor's work on PoL strategy cataloguing and spatial statistical methods informs the taxonomy and composition theory. Advisory role.
- **Ethereum Foundation Use Case Labs** - existing collaboration focused on location proofs and verifiable location-based services to unlock use cases for smart contracts.
- **Open Geospatial Consortium** - standards alignment via STAC integrity/provenance/trust extension and OGC Building Blocks.
- **Kiersten Jowett** - PhD Geospatial Engineering, thesis “Proof of location & the certainty spectrum”

Competency gaps

The primary gap is in formal methods and information theory. The composition theory work - rigorously modeling independence between heterogeneous PoL systems, evaluating candidate formalisms, and defining the credibility vector - requires mathematical expertise beyond the current team. This is the role the quantitative researcher hire is designed to fill. We also see this as a natural point for cross-track collaboration: Track 3.1 (Formal AI Security) and Track 3.3 (Generative Security) teams may bring complementary expertise, and we would actively seek those partnerships through build weeks and programme channels.

Hardware engineering expertise - designing secure peripheral architectures, working with OSNMA receivers, building instrumented test rigs - will be needed in Phase 2 when the work moves from formalization to reference dataset construction and physical deployments.

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